



Course Code: CS362IA

			March 2020				
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Network Programming and Security

Maximum Marks : 100

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer Five full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.

Question No	Question	M	CO	BT
1.1	fork() is called once but returns twice. Explain.	02	1	2
1.2	What happens when the server process receives a FIN segment from the client during TCP connection termination?	02	2	2
1.3	Differentiate between SOCK_STREAM and SOCK_DGRAM socket types with respect to the transport protocols used.	02	1	4
1.4	Why can a concurrent server handle multiple clients simultaneously while an iterative server cannot?	02	1	2
1.5	Why is getaddrinfo() preferred over older functions like gethostbyname() in modern network programming?	02	2	2
1.6	Differentiate between flag options and value options in socket programming.	02	2	1
1.7	A web application uses both IPv4 and IPv6. Which DNS records must be configured, and how does the resolver decide which to use?	02	4	3
1.8	What is the purpose of the Feistel structure?	02	1	2
1.9	Which encryption protocol provides stronger wireless security than TKIP?	02	4	2
1.10	Which HTTP feature confirms that a website's identity has been verified?	02	4	3

Part B

Question No	Question	M	CO	BT
	A multimedia company wants to build a live video conferencing system and the Requirements are • Video • Voice • Text chat • File sharing			
2a	(i) Choose the most suitable transport protocol(s) for each feature with justification. (ii) Explain socket programming concepts required to implement the system. (iii) Explain byte ordering and address conversion functions required for global deployment.	08	2	2
2b	Describe the role of the Transport Layer in computer networks. Compare TCP, UDP, and SCTP based on reliability, connection management, flow control, congestion control, and applications.	08	1	2
3a	Illustrate the normal startup and normal termination of the server using LINUX commands.	10	3	3
3b	Explain the six exec family functions in Unix. Discuss the differences among execl(), execv(), execl(), execve(), execlp(), and execvp() with respect to argument passing, pathname specification, and environment handling.	06	3	2
OR				
4a	With a neat diagram, explain how a concurrent server handles multiple client connections simultaneously. Discuss the responsibilities of the parent and child processes.	10	3	2
4b	Write a code snippet to print the server's assigned port using getsockname() and Print client IP and port after connection using getpeername().	06	3	3
5a	Explain the purpose of the functions getservbyname() and getservbyport in socket programming. Describe how each function works and Identify the role of the servent structure.	08	4	4
5b	Write a code snippet using select() to implement a timeout for a UDP client waiting for a server reply.	08	3	3
OR				
6a	Discuss the working of getsockopt() and setsockopt() system calls in socket programming with suitable code snippet.	08	1	2
6b	Explain the role of resolvers and name servers in the Domain Name System (DNS).	08	3	2

7a	For given 128 bits string 11001010 10110011 01101001 11100010 10101100 01011110 11010010 00110101 11110000 10101010 01010101 10011001 11001100 00110011 11111111 00000000, design a S-Box and produce two 48 bit blocks of 4 bits. Show each step and output string.	08	3	3
	7b Explain working of stream ciphers and Block ciphers with example	08	2	2

OR

8a	Analyze the computational aspects of the RSA algorithm. Explain how key generation, encryption, decryption, and key size affect the computational efficiency of RSA. Compare the computational cost of each phase and justify why RSA is mainly used for key exchange and digital signatures rather than bulk data encryption.	08	4	4
8b	Evaluate where, Alice and Bob wish to establish a shared secret key using the Diffie–Hellman key exchange algorithm with given, Prime number, $p = 23$, Primitive root, $g = 5$, Alice's private key, $a = 6$, Bob's private key, $b = 15$. by determining the following: i) Alice's public key ii) Bob's public key iii) The shared secret key computed by Alice iv) The shared secret key computed by Bob	08	5	5
9a	A multinational company deploys a wireless network for its employees and guests. During a security audit, it is observed that the wireless network uses the default SSID and administrator password, SSID broadcasting is enabled, endpoint devices do not have antivirus protection, communication is not encrypted, and any nearby device can connect to the network. Recommend suitable security measures to secure the wireless network and justify each recommendation.	06	4	4
9b	Explain the complete TLS handshake protocol with a neat sequence diagram.	10	4	2

OR

10a	Analyze the major security threats to web security with respect to integrity, confidentiality, denial of service, and authentication. Discuss their consequences and suitable countermeasures.	10	4	4
10b	Describe the key factors that contribute to the security risks in wireless networks.	06	4	2